Joshua Duplaa

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EXPERIENCE

Software Engineer, Mercor — Feb. 2026 - Present
Seattle, Washington (Remote, Contract)

- Develop solutions for SWE-bench style software engineering tasks used to evaluate AI coding agents, debugging real-world repositories and implementing fixes under strict test constraints.
- Implemented secure HTML sanitization logic to mitigate cross-site scripting (XSS) attacks and built utilities for reproducible data processing and testing in Linux environments.

DevOps Engineer Intern II, ResMed — May 2024 - Aug. 2024
San Diego, California (Remote)

- Aided recovery during the global CrowdStrike outage; restored 25+ Windows servers manually and helped build automation to recover the remaining fleet, reducing downtime during an SEV-1 incident.
- Monitored and supported production multi-cloud infrastructure across AWS and Azure, primarily using AZ Monitor, integrating Kusto (KQL) queries into monitoring pipelines, improving overall server reliability.
- Built a distributed Kubernetes training platform on AKS with Azure DevOps Pipelines for a 12-person team, cutting onboarding time and enabling self-serve cluster deployments.

Data Analyst, Texas Tech IMSE — Oct. 2023 - Jan. 2025
Lubbock, Texas

- Built and maintained 15+ interactive Power BI dashboards for IMSE and College of Engineering.
- Designed and standardized ETL pipelines and data models (Python, Pandas, DAX, SQL), improving accuracy, consistency, and reporting workflows.
- Produced clear technical documentation for dashboards, data models, and reporting pipelines for engineering and finance stakeholders.

DevOps Engineer Intern, ResMed — June 2023 - Sept. 2023
San Diego, California (Remote)

- Automated Skilled Nursing Facility (SNF) server deployments using Ansible, Terraform, and custom scripts in Python, Bash, and PowerShell, enabling repeatable cloud provisioning.
- Built CI/CD pipelines to test, version, and deploy AWS and Azure infrastructure as code with Terraform.
- Supported cloud networking and security configurations, including VPCs, load balancers, firewalls, ACLs, and DNS, strengthening production readiness.

Undergraduate Research Assistant, Texas Tech Applied Vision Laboratory — Aug. 2022 - Jan. 2023
Lubbock, Texas

- Built data pipelines to collect, segment, and preprocess 100+ hours of video for evaluating real-time computer vision models.
- Supported development and evaluation of neural network models using MATLAB and Python, including PyTorch and Mask R-CNN, to analyze model performance on video data.

Calculus II Tutor, Texas Tech TLPDC — Aug. 2022 - Jan. 2023
Lubbock, Texas

EDUCATION

Texas Tech University, Lubbock, Texas — May 2025
Whitacre College of Engineering — *GPA 3.3*
B.S. Computer Science; B.S. Mathematics (Double Degree, Data Science concentration)
Certifications: *AWS Cloud Practitioner CLF-C01, Azure Fundamentals AZ-900*

SKILLS

Networks and Cloud Infrastructure: AWS, Azure, GCP, Terraform, Ansible, Docker, Kubernetes, CI/CD (GitHub Actions, GitLab, Jenkins, Azure DevOps), CloudWatch, AZ Monitor, Prometheus, Grafana.
Programming Languages: Python, Bash, PowerShell, JavaScript/TypeScript, SQL, C/C++, Rust, R, C#, Java, MATLAB.
Web: Node.js, ASP.NET, Django, Flask, FastAPI, REST APIs, BeautifulSoup, React, Next.js, HTML/CSS.
Data & AI: ETL pipelines and data visualization with Pandas & SQL; Fine Tuning and RAG pipelines; TensorFlow, PyTorch; LLM integration and leveraging AI tools. Sharded datasets and SQL Databases to support scalable ML training workflows.
Spoken Languages: Native in English and Portuguese; Proficient in Spanish, French, and German.